A Comparative Study of Input Port and Crossbar Configurations in NoC Router Microarchitectures

Sneha N Ved[1] and Aparna Arya[2] and Ankit Bhange[2] and Joycee Mekie[3]

***Abstract*— Network-on-Chip (NoC) has emerged as a promising solution as an on-chip interconnect for multi-cores. Most of the research in NoCs revolves around router microarchitecture for power efficiency and performance and in algorithms for efficient data transfer. While many works in the literature discuss the impact of buffer organization on the performance in terms of latency, in this work we evaluate the impact of buffer organization with area and power consumption in the router. In this experimental study, we implement *36* different router architectures with different flit sizes, input port and crossbar configurations. All these router architectures are implemented, synthesized and validated on the 28nm Xilinx Kintex KC705 FPGA hardware for comparison. We have also synthesized these routers using the 1P8M UMC65nm standard cell technology on Cadence ASIC flow. We report that for the same number of buffers, the dynamic power and area can vary by up to 67% based on the configuration. We also find that increasing the number of buffers does not increase area and power by the same proportion. Further we report that the crossbar configuration can be significantly improved by designing routing specific crossbars.**

I. Introduction

As the number of cores on-chip are growing, greater emphasis on efficient inter-core on-chip communication is required. Network-on-Chip (NoC) is a suite of infrastructure and protocols to carry out inter-core communication. Typically, an NoC consists of the routers (R) and network interfaces (NI) that connect the cores (C). The routers are physically connected to one another over a topology.

What would be the ideal buffer organization for a given router design? This is a question oft faced by router designers. The choice of buffer organization is usually based on the performance required. However, with area occupied and power consumption as important metrics of router evaluation, choice of appropriate buffer organization and crossbar architecture becomes critical.

Most of these router architectures have been proposed with focus on network parameters such as performance, quality of service, packet latency, sustainable injection rate etc. However, there is little done to evaluate different router architectures for area and power consumption with respect to the buffering schemes, especially in the hardware implementation. In this work, we assess the impact of flit size, crossbar design and input port configurations on the power and area of the router. Based on different flit sizes, input port organizations and crossbar configurations, we implement 36 different router architectures. We validate the implemented routers on 28nm Xilinx Kintex KC705 FPGA and evaluate their power and area using the Cadence ASIC design flow for UMC 65nm 1P8M technology.

A. Our Contributions

In this work, we implemented routers with different input port organizations for 38-bit and 68-bit wide flits. We assess the input port configurations with constant storage (constant amount of total buffering) or with constant flit depth (constant amount of flit storage per VC). We also explore crossbar architectures in association with the routing algorithm from the perspective of area and power optimization. We implement two customizations of the baseline crossbar for dimension order-XY routing algorithm in a 5-port router.

Based on the input port configurations and routing algorithm specific crossbars, 36 different router architectures are implemented. The router architectures were synthesized for 28nm Xilinx Kintex KC705 FPGA for hardware validation and also for the Cadence ASIC design flow for UMC 65nm 1P8M technology to evaluate for power and area. The key contributions of this work are:

1) We show that input port configurations with minimum number of virtual channels is beneficial in terms of area and power
2) We also show that the increase in the input port area and power is is sub-linear to the increase in the buffering at the input port
3) We demonstrate that routing algorithm specific crossbars are more area and power efficient than the generic baseline crossbar.

We report that the area occupied and the power consumed vary significantly over different router architectures. However, generally speaking, routers designed using dimension order-XY routing customized hierarchical router (shown in Fig. 5(c)) occupy lesser area and consume lesser power than their counterparts implemented using other crossbars. We observe that the increase in area and power consumption does not scale linearly with increase in buffers or number of VCs in routers designed using either of the crossbars. This is because in addition to the area occupied and power consumed by the buffers in the input ports, multiplexers, demultiplexers and associated logic also contribute to the area and power of the router. Thus, while the number of buffers in the input port configuration with 4 VCs, each capable of storing up to 4

*This work was funded by Intel India under Intel PhD Fellowship Programme

[1]Sneha N Ved is a PhD candidate with Indian Institute of Technology Gandhinagar, India

[2]Aparna Arya and Ankit Bhange are B.Tech students at Indian Institute of Technology Gandhinagar, India

[3]Prof. Joycee Mekie is a member of faculty at Indian Institute of Technology Gandhinagar, India

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL. Downloaded on August 24,2026 at 09:26:25 UTC from IEEE Xplore. Restrictions apply.

978-1-5090-2797-2/17/$31.00 ©2017 IEEE

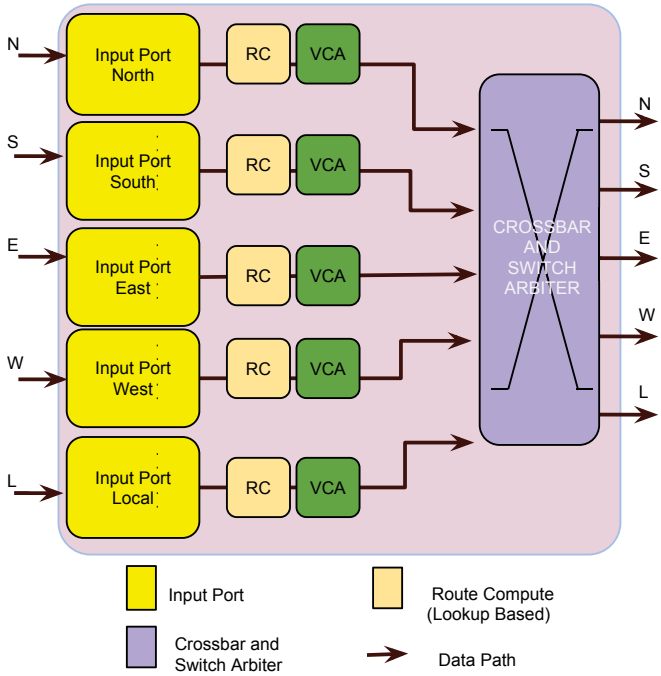


Fig. 1: Baseline Router

flits is twice that in the input port configuration with 4 VCs, each capable of storing 2 flits, it occupies only about 1.44 times the area and consumes only about 1.23 times dynamic power. This observation is of value while designing an NoC with a desired performance, area and power budget.

II. Related Work

Existing work in NoC encompasses an entire spectrum of work from bufferless routing [1][2][3][4] to deeply buffered routers [5][6][7][8]. Several works discuss use of virtual channels [9][10] for improvement in network performance. Several router architectures favoring specific topologies [11] have been proposed in the literature. Router power and energy have also been in focus in the NoC research. Methods such as power gating[12][13][14] and frequency scaling[15] have been used to reduce the power dissipation in the routers. While there is a significant amount of work in the literature proposing newer architecture, very little is done to compare them on the same platform. This work is a step towards quantifiable comparison of different router microarchitectures.

III. Router Microarchitecture

Fig. 1 shows the architecture of a typical router. This is a 3-stage pipeline router with Input Buffering, Route Compute and Crossbar Traversal stages. The incoming data flit is queued until all the preceding flits in the same FIFO are processed. The input port is typically organized as a set of FIFOs called virtual channels (VC) [9]. Once read from the virtual channel, the output port for the data flit is evaluated at the Route Compute (RC) stage. In the same pipeline stage, in addition to picking the output port, the Virtual Channel Allocator (VCA) also allocates one of the virtual channels at the input port of the subsequent router. After the allocation of the output port and the downstream VC, the data flit passes through the crossbar switch which physically connects the input port to the output port. Thus, in the third pipeline stage, the flit physically traverses from the input port to the output port. For this work, we have considered routers with a flit width of 38 bits and 68 bits respectively.

A. Input Port

Input port is organized as a set of FIFOs called virtual channels (VC). While the advantage of having buffered input is that it allows larger number of in-flight packets and can be used for deadlock recovery, message classification etc, buffers occupy a significant amount of area and consume power [1].In a large majority of cases, estimating the depth of the buffers at design time is a challenge owing to a large variety of applications that need to be run on the system. In a VC based system, the appropriate number of VCs depends on several factors including data access patterns in target applications, number of message classes to be supported, QoS desired, area-power budget etc. Reconfigurable buffering schemes have been proposed to attend to the issue of finding appropriate number of VCs required in the NoC. While these address the problem of deciding the input port organization as seen by the router at run-time, they come with associated power and area overheads for the reconfiguration logic. In this work, we try to quantify the area and power occupied by different buffer organizations. We implement routers supporting 38-bit and 68-bit flit formats in configuration having upto 8 and 4 virtual channels respectively. The different input port configurations implemented on FPGA and in Cadence ASIC tool flow are listed in Tab. I.

To understand the effect of area consumption and power utilization changes across different input port configurations, the assessed configurations were chosen to have either constant amount of flit storage or constant amount of VC depth. To account for various architectures with constant flit storage capacity, total input buffer storage is fixed, say s. The available v VCs are then assigned to have depth d such that $s = v \times d$. To derive different input port configurations with constant depth, a depth d is fixed and routers with different number of VCs are implemented, where each VC can store upto d flits at any given time.

B. Route Compute (RC) and Virtual Channel Allocation (VCA) Unit

We have implemented a table look-up based route compute using dimension order-XY routing algorithm. XY routing is chosen due to its simple implementation and delivery guarantees. XY-routing is deadlock and livelock free, deterministic routing algorithm. RC takes the destination address as the input and generates the port id of the appropriate output port as the output. The port id of the output port is then given to the VCA to allocate a suitable virtual channel to the flit in the subsequent router. VCA accepts the assigned output port id as the input to generate the VC number. In case the flit is assigned a VC other than its current VC, the VC number field in the flit is modified to the new value.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL. Downloaded on August 24,2026 at 09:26:25 UTC from IEEE Xplore. Restrictions apply.

C. Crossbar

We have implemented three variants of the crossbar architecture for both 38-bit and 68-bit routers as shown in Figs. 2, 3, 4 respectively. Baseline crossbar from Fig. 2 is a generic complete crossbar which connects all the input ports to all output ports other than itself. It can be used for any 5-port router irrespective of the routing algorithm. However, in case of dimension order-XY routing, Y dimension to X dimension turns are prohibited. Reduced baseline crossbar as shown in Fig. 3 eliminates the connections prohibited by the dimension order-XY routing giving a more customized compact crossbar than the baseline crossbar.

In dimension order-XY routing, each flit moves from X-dimension to the Y-dimension atmost once. Thus, in a majority of the in-flight routers, the passing data flit does not need to get access to all the output ports. This makes it possible to segment the conventional generic Baseline router into X-dimension sub-router and Y-dimension sub-router as shown in Fig. 4. This splitting of the router requires addition of a three multiplexers, which were neither present in baseline (Fig. 4) nor reduced baseline (Fig. 3) crossbars.

IV. Experimental Results

We implemented 36 routers using the input port and crossbar configurations discussed in section III. These routers have been implemented in Verilog and tested on 28nm Xilinx Kintex KC705. Fig. 6 shows the RTL schematic of a router with hierarchical crossbar for 38-bit wide flit as generated for the FPGA implementation of the router. This RTL schematic is generated on synthesis of the design on Xilinx tool flow. The inset figures are the schematics of the input port configuration and the crossbar microarchitecture used in the router. Fig. 7 is the RTL schematic generated on putting the router through the Cadence ASIC tool flow using UMC 65nm 8M standard cell technology. As indicated in the figure, the router layed out in an area of approximately 165×362 on a die of 400×400 um. The input port for the router shown in Fig. 6 has 4 VCs each of depth 4. For area and power analysis, these routers were synthesized using the Cadence ASIC tool flow. Table I lists the area and power of the implemented routers. Fig. 8 shows a snapshot of the 38-bit router with input port configuration as 4 virtual channels, each capable of holding up to 4 flits and a hierarchical crossbar. The credit signals are the inputs to the virtual channel allocator unit of the router.

Discussion

It is observed that the change in the area and power consumption is not in direct proportion with the change in the buffering. Consider the case of 4VC depth 2 and 4VC depth 4 input port architectures of the 38-bit router with baseline crossbar. While the number of buffers doubles, from 8 buffers in the former to 16 buffers in the latter, the area increases only by 1.75 times and the dynamic power by 1.43 times. Consider the case of 4VC 2 and 2VC 4 configurations of input port organization in 38-bit router with baseline crossbar. While the number of buffers in both the cases is the same i.e. 8 buffers, we observe that 4VC 2 configuration

Router Type	Area	Static Power	Dynamic Power
38-bit Baseline			
1 VC - Depth 4	0.013	3.83	1.11
2 VC - Depth 4	0.028	8.41	2.55
4 VC - Depth 4	0.049	14.83	3.78
8 VC - Depth 4	0.050	14.38	5.09
68-bit Baseline			
1 VC - Depth 4	0.025	7.32	2.20
2 VC - Depth 4	0.036	10.64	3.23
4 VC - Depth 4	0.072	21.79	5.41
38-bit Reduced			
1 VC - Depth 4	0.014	3.99	1.14
2 VC - Depth 4	0.024	7.03	2.32
4 VC - Depth 4	0.047	14.07	3.71
8 VC - Depth 4	0.049	14.25	5.20
68-bit Reduced			
1 VC - Depth 4	0.022	6.63	1.96
2 VC - Depth 4	0.035	10.45	3.22
4 VC - Depth 4	0.095	27.47	4.07
38-bit Hierarchical			
1 VC - Depth 4	0.017	5.00	1.55
2 VC - Depth 4	0.017	4.73	1.90
4 VC - Depth 4	0.043	12.97	3.45
8 VC - Depth 4	0.050	14.26	5.14
68-bit Hierarchical			
1 VC - Depth 4	0.015	4.03	1.58
2 VC - Depth 4	0.037	10.75	2.51
4 VC - Depth 4	0.088	26.80	6.37

(a) Constant Depth Configurations (Depth = 4)

Router Type	Area	Static Power	Dynamic Power
38-bit Baseline			
1 VC - Depth 8	0.023	6.90	1.85
2 VC - Depth 4	0.028	8.41	2.55
4 VC - Depth 2	0.034	10.13	3.07
8 VC - Depth 1	0.049	14.05	5.15
68-bit Baseline			
1 VC - Depth 4	0.025	7.32	2.20
2 VC - Depth 2	0.031	8.95	3.25
4 VC - Depth 1	0.037	11.04	3.55
38-bit Reduced			
1 VC - Depth 8	0.020	6.11	1.63
2 VC - Depth 4	0.024	7.03	2.32
4 VC - Depth 2	0.032	9.37	3.03
8 VC - Depth 1	0.047	13.41	5.11
68-bit Reduced			
1 VC - Depth 4	0.022	6.63	1.96
2 VC - Depth 2	0.035	10.55	3.52
4 VC - Depth 1	0.037	10.85	3.54
38-bit Hierarchical			
1 VC - Depth 8	0.024	7.10	1.94
2 VC - Depth 4	0.017	4.73	1.90
4 VC - Depth 2	0.030	8.80	2.86
8 VC - Depth 1	0.050	14.30	5.10
68-bit Hierarchical			
1 VC - Depth 4	0.015	4.03	1.58
2 VC - Depth 2	0.037	10.92	3.67
4 VC - Depth 1	0.043	12.98	3.82

(b) Constant Storage Configurations

TABLE I: Area and power comparison of router architectures

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL. Downloaded on August 24,2026 at 09:26:25 UTC from IEEE Xplore. Restrictions apply.

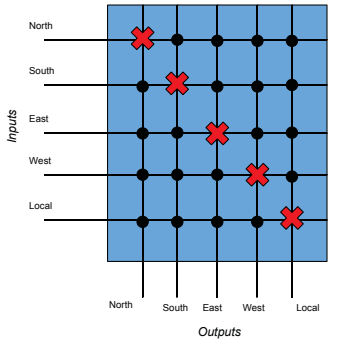


Fig. 2: Baseline Xbar

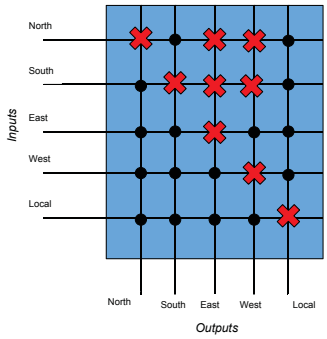


Fig. 3: Reduced Baseline Xbar

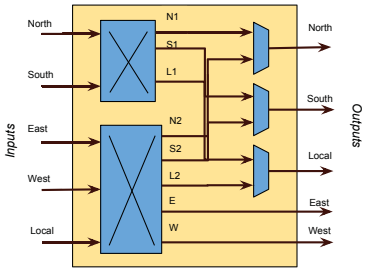


Fig. 4: Hierarchical Xbar

Fig. 5: Crossbar Microarchitectures

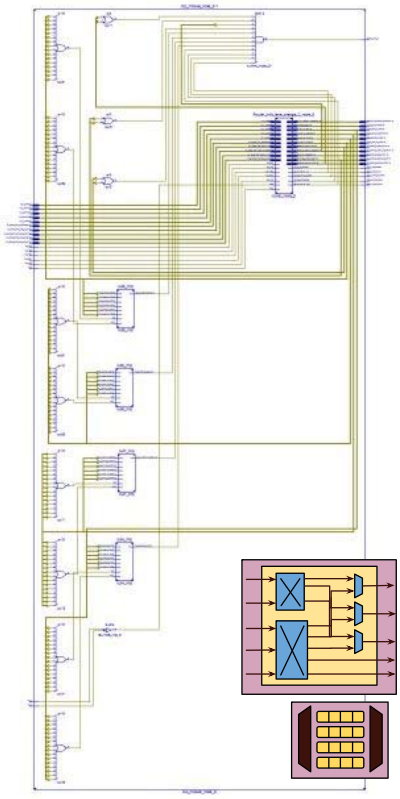


Fig. 6: RTL schematic of 38-bit router with 4 VCs each of depth 4 and hierarchical crossbar (Inset: Input port configuration and crossbar used)

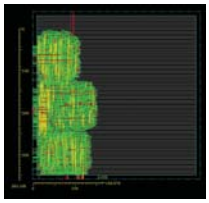


Fig. 7: ASIC schematic of 38-bit router with 4 VCs each of depth 4 and Baseline crossbar

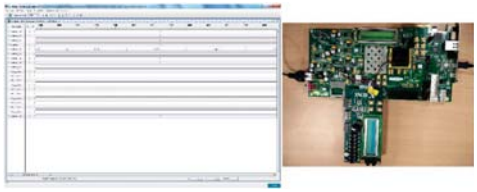


Fig. 8: Chipscope trace of FPGA implementation of 38-bit 4VC depth 4 router with hierarchical crossbar (Inset: Experimental setup used)

occupies about 1.2 times the area of the 2VC 4 router and also consumes about 1.2 times the power. This is due to the wider multiplexer/demultiplexer and control logic required for the 4 VC arrangement than the 2 VC arrangement. Thus, when deciding for an appropriate input port configuration, in addition to the buffering, the complexity of the associated control logic also plays a significant role in determining the area and power overheads incurred.

The following are the observations from the table:

1) Static and dynamic power consumption in routers can vary upto 44% between the different router configurations
2) In general, the variation between the different input port and crossbar combination is more profound in case of 68-bit routers
3) In general, across different crossbar architectures, in routers designed using hierarchical crossbars, the variation observed in area and power over different input port configurations is the least.

Fig. 9 and Fig. 10 compare the area and the dynamic power of 38-bit routers for different input port and crossbar configurations. It is clear from the plots that in a majority of the cases, routers with hierarchical crossbars outperform the baseline and the reduced crossbar routers. However, in some cases, for instance, in case of the 38-bit routers with 8 VCs each of depth 4 input port configuration, the amount of dynamic power consumed across routers with different crossbars is negligible as opposed to the 2 VC with depth 4 configuration.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL. Downloaded on August 24,2026 at 09:26:25 UTC from IEEE Xplore. Restrictions apply.

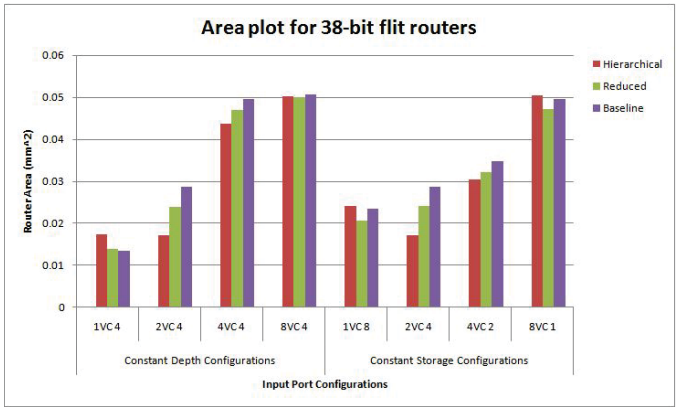


Fig. 9: Area comparison of 38-bit routers with different input port and crosbar configurations

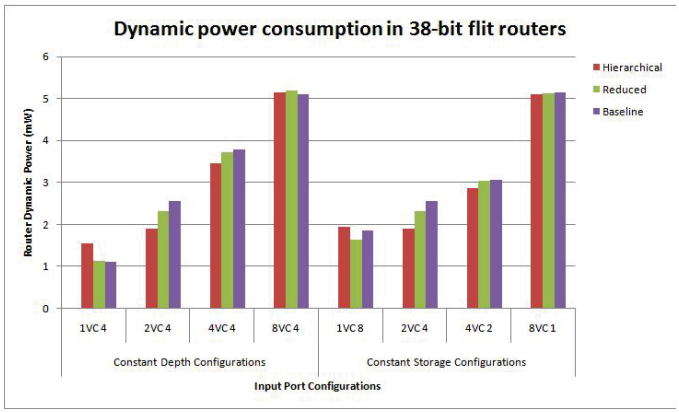


Fig. 10: Power comparison of 38-bit routers with different input port and crossbar configurations

V. Conclusion

While a large part of the literature aims at developing more performance efficient routers, very little is done to help designer choose an optimal router architecture from the perspective of area and power. In this work, we assess router architectures different input port and crossbar organizations for area and power. All the designs have been implemented in Verilog, synthesized for and validated on 28nm Xilinx Kintex KC705 FPGA. The designs are put through the Cadence ASIC design flow using UMC 65nm 8M standard cell technology library for power and area estimates. We have implemented 36 38-bit wide and 68-bit wide routers with different input port and crossbar configurations. The input port configurations are chosen such that either the total amount of storage across all the virtual channels remains constant or the depth of all the virtual channels in the input port is constant. In addition to a generic baseline crossbar, two routing algorithm specific crossbars have been implemented. Routing algorithm specific crossbars have been designed for a router in a 2-D mesh topology that uses dimension order-XY routing. Standard Baseline (Fig. 2) crossbar and dimension order-XY routing specific reduced baseline (Fig. 3) and hierarchical (Fig. 4) crossbars have been implemented for this study. The area and power dissipation information as shown in table. I. Fig. 10 and Fig. 9 compare the router architectures for the dynamic power consumption and area occupied respectively. It is clear from the discussion in section IV that input port configurations with lesser number of VCs are more area and power efficient than their counterparts with more VCs, even when the amount of buffering is the same. We have found that based on the number of VCs for a fixed amount of buffering, the input port area and dynamic power can vary by upto 67% and 68% respectively.

References

[1] T. Moscibroda and O. Mutlu, "A case for bufferless routing in on-chip networks," in *Proceedings of the 36th Annual International Symposium on Computer Architecture*, ser. ISCA '09, 2009.

[2] G. R. Jonna, J. Jose, R. Radhakrishnan, and M. Mutyam, "Minimally buffered single-cycle deflection router," in *Proceedings of the Conference on Design, Automation & Test in Europe*, ser. DATE '14, 2014.

[3] C. Fallin, G. Nazario, X. Yu, K. K. Chang, R. Ausavarungnirun, and O. Mutlu, "Minbd: Minimally-buffered deflection routing for energy-efficient interconnect," in *2012 Sixth IEEE/ACM International Symposium on Networks-on-Chip (NoCS), Copenhagen, Denmark, 9-11 May, 2012*, 2012, pp. 1–10.

[4] H. Kim, C. Kim, M.-G. Kim, K. Won, and J.-H. Kim, "Extending bufferless on-chip networks to high-throughput workloads," in *Networks-on-Chip (NoCS), 2014 Eighth IEEE/ACM International Symposium on*. IEEE, 2014, pp. 9–16.

[5] I. Seitanidis, A. Psarras, G. Dimitrakopoulos, and C. Nicopoulos, "Elastistore: An elastic buffer architecture for network-on-chip routers," in *Proceedings of the Conference on Design, Automation & Test in Europe*, ser. DATE '14, 2014, pp. 240:1–240:6.

[6] A. K. Kodi, A. Louri, and J. M. Wang, "Design of energy-efficient channel buffers with router bypassing for network-on-chips (nocs)." in *ISQED*. IEEE Computer Society, 2009, pp. 826–832.

[7] C. V. Thiem and S. Oyanagi, "An input buffer architecture for on-chip routers." in *ICNC*. IEEE Computer Society, 2011, pp. 280–283.

[8] S. M. Hassan and S. Yalamanchili, "Centralized buffer router: A low latency, low power router for high radix nocs." in *NOCS*. IEEE, 2013, pp. 1–8.

[9] W. J. Dally, "Virtual-channel flow control," in *Proceedings of the 17th Annual International Symposium on Computer Architecture*, ser. ISCA '90, 1990.

[10] R. Mullins, A. West, and S. Moore, "Low-latency virtual-channel routers for on-chip networks," in *Proceedings of the 31st Annual International Symposium on Computer Architecture*, ser. ISCA '04, 2004.

[11] J. Kim and H. Kim, "Router microarchitecture and scalability of ring topology in on-chip networks," in *Proceedings of the 2Nd International Workshop on Network on Chip Architectures*, ser. NoCArc '09, 2009, pp. 5–10.

[12] A. Mirhosseini, M. Sadrosadati, A. Fakhrzadehgan, M. Modarressi, and H. Sarbazi-Azad, "An energy-efficient virtual channel power-gating mechanism for on-chip networks," in *Proceedings of the 2015 Design, Automation & Test in Europe Conference & Exhibition*, ser. DATE '15, 2015.

[13] D. H. Woo, J. B. Fryman, A. D. Knies, and H.-H. S. Lee, "Chameleon: Virtualizing idle acceleration cores of a heterogeneous multicore processor for caching and prefetching," *ACM Trans. Archit. Code Optim.*

[14] A. Mirhosseini, M. Sadrosadati, A. Fakhrzadehgan, M. Modarressi, and H. Sarbazi-Azad, "An energy-efficient virtual channel power-gating mechanism for on-chip networks," in *Proceedings of the 2015 Design, Automation & Test in Europe Conference & Exhibition*, ser. DATE '15, 2015.

[15] P. Zhou, J. Yin, A. Zhai, and S. S. Sapatnekar, "Noc frequency scaling with flexible-pipeline routers." IEEE/ACM, 2011.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL. Downloaded on August 24,2026 at 09:26:25 UTC from IEEE Xplore. Restrictions apply.